

ICS 604: Applied Data Science

Description: Introduces foundational principles of data science, emphasizing elementary concepts and essential tools integral to the data scientist's role from data wrangling to foundational tools for modeling data, without requiring knowledge of linear algebra or calculus.

Prerequisites: This course requires to have basic programming skills in Python. Prior knowledge of data science methods is helpful but not required.

Class Meetings: MW 12:00 PM – 1:15 PM in POST 325 and virtually via Zoom

Instructor: Kyungim Baek

- Office: POST 305-C
- Office hours: Wednesdays 1:30 PM – 3:00 PM, or by appointment
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Teaching Assistants:

- Ethan Chung
 - Office: POST 303-B
 - E-mail: echung32 (AT) hawaii.edu
- Akib Sadmanee
 - Office: POST 306-A
 - E-mail: sadmanee (AT) Hawaii.edu
- TA office hours will be held by appointment. Please contact Ethan and/or Akib to schedule a session. You may also post questions on the course Slack channel.

Course Learning Objectives

To teach students:

- Python scientific stack for data munging and processing
- Statistical and probabilistic concepts and methodologies underlying data science
- Popular techniques for modeling data
- Common Python data visualization libraries

Student Learning Outcomes

- Familiarity with popular data science IDEs such as Jupyter Notebooks
- Advanced skills in Pandas and data processing of tabular and time series data
- Understand and apply the principles of simulations
- Understand and apply fundamental probability distributions
- Understand and apply methods for parameter estimation
- Understand and apply fundamental hypothesis testing methods
- Understand and apply simple concepts for time series data analysis
- Understand basic regression and classification methods
- Understand the training, validation, and testing phases of machine learning project

Textbook: There is no required textbook for this course. All course materials will be made available online via Lamaku.

Reference for data wrangling with Pandas:

- McKinney, Wes. Python for data analysis, 3rd Edition, O'Reilly Media, Inc., ([Open edition](#))

Tentative Schedule

This is a tentative outline. It is subject to change as the class progresses.

- Week 1: Course logistics, Introduction to data science and Pandas
- Week 2: *No class on Monday (MLK Jr. Day)*, Data exploration

- Week 3: Arithmetic operations, Data alignment, Data wrangling with Pandas: Data preparation and cleaning
- Week 4: Data wrangling with Pandas: Split-apply-combine, Hierarchical indexes, Join and combine data
- Week 5: *Class cancelled on Monday (campus closure)*, Data visualization with Matplotlib
- Week 6: *No class on Monday (President's Day)*, Introduction to probability
- Week 7: Probability distributions, Kernel density estimation (KDE)
- Week 8: Parameter estimation and simulation (bootstrap, MLE, approximate Bayes)
- Week 9: Hypothesis testing
- Week 10: **** Spring Recess ****
- Week 11: Hypothesis testing
- Week 12: Correlation, Linear and non-linear regression
- Week 13: Linear and non-linear regression
- Week 14: Time series analysis
- Week 15: Clustering (hierarchical, K-means, GMMs)
- Week 16: Machine learning (ML) intro, Popular ML models
- Week 17: Model validation

Grading

- Grading criteria and weights:
 - Homework assignments: **55%**
 - Class participation: **10%**
 - Final project: **35%**
 - Project proposal (7%): **Due Monday, March 23**
 - Peer feedback on project proposals (8%): **Due Monday, March 30**
 - Final report and code (20%): **Due Monday, May 11**
- Grading scale:
 - Point totals will be used as-is; **no rounding will be applied.**

Letter Grade	Grade Points		
As	90-91.99 A–	92-97.99 A	98-100 A+
Bs	80-81.99 B–	82-87.99 B	88-89.9 B+
Cs	70-71.99 C–	72-77.99 C	78-79.9 C+
Ds	60-61.99 D–	62-67.99 D	68-69.9 D+
F	0-59.99		

Course Policies

- There is no extra credit in this course.
- Late work is not accepted without prior approval, which is only granted in unavoidable circumstances. Student should contact the instructor if any significant delays arise.
- Students will be notified within a reasonable time of any necessary changes to grading criteria.
- Photographing or making electronic video and/or audio recording during class are not permitted.
- All course materials will be posted on Lamakū. Please do not post or upload any of the course materials to other websites.
- Attendance: Students who are enrolled in this course but never attend will be flagged by the

course instructor for non-participation before the last day to add/drop (for 100% tuition refund) deadline. Flagged students will be administratively dropped by the Office of the Registrar. Any changes to a student's enrollment status may affect financial aid eligibility and can result in the return of some or all of federal student financial aid.

Academic integrity and ethical behavior

- Cheating, plagiarism, or other forms of academic dishonesty are not permitted within this course and are prohibited within the System-wide Student Conduct Code (EP 7.208). Examples include: fabrication, facilitation, cheating, plagiarism, and use of improper materials. Any incident of suspected academic dishonesty will be reported to the Office of Judicial Affairs for review and possible adjudication.
- All occurrences of academic dishonesty will result in a zero grade for the assignment/exam in question the first time it is detected, and a grade of F for the course for any subsequent instance. **This policy will be applied to those who copy other's work and who allow their work to be copied.** You can discuss strategies or general approaches to solving assignments with other students, however collaboration on solutions/codes, sharing or copying of solutions/codes is not allowed. You should write up your own answers/codes and should not show your work to other students.
- **Use of Artificial Intelligence:** AI tools may be used in a supportive way — for example, to help generate code, diagnose errors, or clarify concepts. However, you are responsible for carefully reviewing, testing, and revising any AI-generated output to ensure that it is correct, meets the requirements of the course activity, and reflects your own understanding. You should not treat AI tools as “black boxes” that produce ready-made, copy-and-paste solutions without meaningful engagement with the underlying logic, as this can lead to overreliance on AI and hinder learning and genuine skill development.
- Using code available online or generated by online tools is not allowed in general. Individual cases should be discussed with the instructor for exception.
- You may consult public literature (books, articles, etc.) for information, but you must cite each source of ideas you adopt.

University Resources

- **Disability Accommodations.** If you have a disability and related access needs, please contact the KOKUA program (UH Disabled Student Services Office) at 956-7511, KOKUA@hawaii.edu, or go to Room 013 in the Queen Lili'uokalani Center for Student Services. Please know that I will work with you and KOKUA to meet your access needs based on disability documentation. KOKUA's services are confidential and offered free of charge.
- **The Counseling and Student Development Center (CSDC)**, a division of the Office of Student Affairs, has as its overall mission the provision of high-quality services to promote the academic success and well-rounded development of its students. CSDC serves the mental health and career counseling needs of students, faculty, and staff at UHM. The center can be reached at (808) 956-7927 in room 312 of the Queen Lili'uokalani Center for Student Services. You can also visit their website at <http://manoa.hawaii.edu/counseling/>.
- **Office of Title IX:** (808) 956-2299 / t9uhm@hawaii.edu / <https://manoa.hawaii.edu/titleix/>
- **Department of Public Safety:** (808)956-6911 (Emergency) / (808)956-8211 (non-Emergency) <https://manoa.hawaii.edu/dps/>
- **UH System Basic Needs** include food and housing, childcare, mental health, financial resources and transportation, among others. Student basic needs security is critical for

ensuring strong academic performance, persistence and graduation and overall student well being. If you or someone you know are experiencing basic needs insecurity, please see the following resources: <https://www.hawaii.edu/student-basic-needs/resources/manoa/>.

- **Student Success Resources:** The Division of Student Success (DSS) houses student support services to build success inside and outside the classroom. If you want learning assistance, academic advising, career resources and guidance, counseling, family and relationship support, identity-based support, services for underrepresented groups, health and wellness services, opportunities for leadership growth, and community engagement, you will find this and more in DSS, (808) 956-3290, <https://manoa.hawaii.edu/studentsuccess/departments/>.